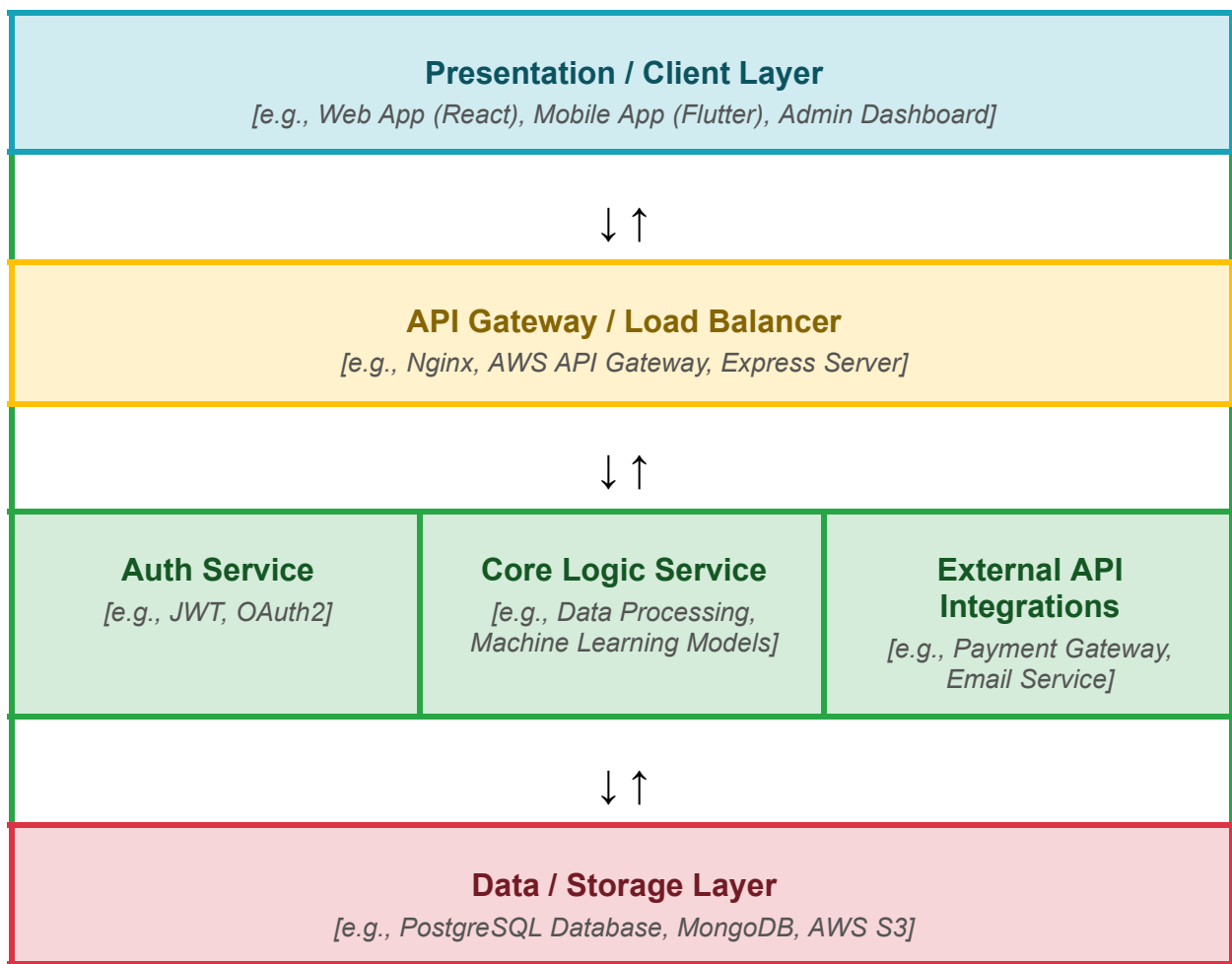


Solution Architecture

Date	05 July 2026
Team ID	SWTID-2026-8308
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	<i>Provides a user-friendly interface for farmers and administrators to monitor crops, view recommendations, weather updates, and analytics.</i>	<i>HTML, CSS, JavaScript, React.js (or Bootstrap)</i>
API Gateway	<i>Acts as the communication bridge between the frontend and backend. It receives user requests and securely routes them to the appropriate services.</i>	<i>Node.js, Express.js, REST API</i>

Microservices	<i>Processes crop recommendations, weather analysis, yield prediction, fertilizer suggestions, and production optimization using AI/ML models.</i>	<i>Python, Flask, Scikitlearn, Pandas</i>
Database	Stores farmer details, crop information, soil data, weather records, prediction results, and historical production data securely.	MySQL (or MongoDB)